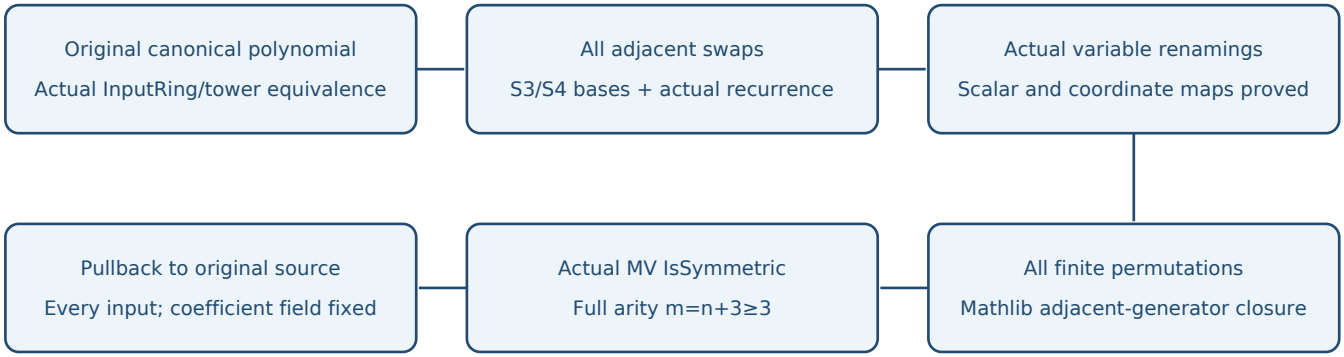
Full symmetry of the actual canonical polynomials

Proved component. For every field K, actual Weierstrass curve W and stage n, the existing recursivePolynomial W n is invariant under every permutation of its m=n+3 inputs. Its literal multivariate image is IsSymmetric. Signed standing: unverified. [pcc:unverified]

Actual action. The pullback permutation sends inputVariable(i) to inputVariable(σ(i)), fixes every C(coefficientEmbedding n a), and identifies outer coordinate0 with X. Explicit ring equivalences connect this action and the multivariate polynomial to the original canonical source. No symmetry or recurrence is assumed. [pcc:unverified]



Induction. S3 gives the arity3 base. Actual-curve S4 middle symmetry gives the arity4 case. The proved source-ordered S4 block handles the newest-three boundary swap at later stages. Older swaps pass through the actual resultant coefficient maps and the prior stage's invariance. This proves every adjacent swap at every stage. [pcc:unverified]

Global bridge. An explicit multivariate/tower equivalence maps every scalar and variable correctly. Each tower adjacent action is proved equal to the corresponding Fin-index variable renaming. Mathlib mclosure_swap_castSucc_succ says these swaps generate the finite symmetric group. The resulting IsSymmetric theorem is pulled back to the actual original polynomial. [pcc:unverified]

Scope and control. No nonsingularity or characteristic restriction is required for this polynomial identity. The inherited rational independent-b S4 counterexample b=(0,0,0,1), inputs(0,1,2,3), refutes an unrestricted coefficient extension; the parent Lean proof rules out a rational curve with those b-coefficients. No qualified source theorem is refuted. General splits k≥3 and higher absolute irreducibility remain open. [pcc:unverified]

Checks. Whole-project gate 90/0failures/0trusted. Lean4.35.0 / Mathlib56a00585…. Only standard foundations; no holes, custom axioms, native evaluation, raised limits, global elaboration flags or waivers. Tower is a transparent abbreviation for actual polynomial rings. See sources.json, proof-map.json and evidence/ for exact bindings, complete custom source pins, failed attempts and whole-gate records. [pcc:unverified]

Sources. Original Semaev2004 Theorem1/p2; Kosters–Yeo2015 Section2/p3 general formula and recursion; actual canonical/transport/S4 modules and the immutable source-order packet; Mathlib finite permutation closure and MV rename/equivalence APIs. KB unavailable; bounded fallback, no novelty claim. No native experiment, paid compute or agent upload. The all-fourteen audit retains its remaining obligations. [pcc:unverified]